

EXPLORATORY DATA ANALYSIS OF
'RAW HOUSE DATA'
SUMMARY
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Given the data 'raw_house_data.csv' I have performed an EDA. I have removed outliers and filled in missing data as described further below. In addition, I have also plotted histograms, pair plots, heatmap, and box plots with the columns containing numeric data. I renamed the columns to more 'user-friendly' labels. I then saved this cleaned data to a new file. This presentation is directed towards other data scientist within my team who will be working on this data.

- Used CoLab with Python and imported pandas, Seaborn, and Matplotlib Pyplot libraries.
- Checked for duplicates in the MLS column but none were found.
- Removed 15 outliers from the following (with amounts):

lot_acres	4
taxes	2
bedrooms	3
bathrooms	1
HOA	5

- Replaced nulls from the following (with amounts and action):

lot_acres	10	Replaced with mean.
bathrooms	6	Replaced with mean.
sqrt_ft	54	Replaced with mean.
garage	5	Replaced with 0.
kitchen_features	33	Replaced with 'None'.
floor_coverings	1	Replaced with 'None'.
HOA	557	Replaced with -1.

- Replaced zeros from the following (with amounts and action):

lot_acres	35	Replaced with mean.
year_built	5	Replaced with mean.

- After this cleaning of data I performed a 'drop_duplicates' action but none were dropped.
- Saved cleaned data to file 'raw_house_data_cleaned.csv'.
- Renamed columns to more legible phrases for plots (ie, 'sqrt_ft' became 'Square Footage').
- Created histograms, pair plots, heatmap, and boxplots for the following *except* those in bold: **MLS**, Sold Price, ZIP Code, Lot Acres, Taxes, Year Built, Bedrooms, Bathrooms, Square Footage, Garage, **Kitchen Features**, Fireplaces, **Floor Covering**, HOA
- Cleaned data saved to 'raw_house_data_clean.csv'.